

Roll: 220125

GitHub Profile: <https://github.com/dibakar1612>

GitHub Repo: <https://github.com/dibakar1612/Linear-Logistic-Regression/>

Collab File Links:

1. Linear Regression:
https://colab.research.google.com/drive/19Y7XAQdKlockI919qRCKL5NTIjF83W0L?usp=drive_link
2. Logistic Regression:
https://colab.research.google.com/drive/11a7MUzZjidz_ShJ8Y1MUt_RchHt7yXBc?usp=drive_link

Dataset Links:

1. Student Performance Dataset (Used for linear regression):
[Drive link](#)
[Kaggle link](#)
2. Titanic Dataset (Used for logistic regression):
[Drive link](#)
[Kaggle link](#)

Screenshots:

1. Linear Regression

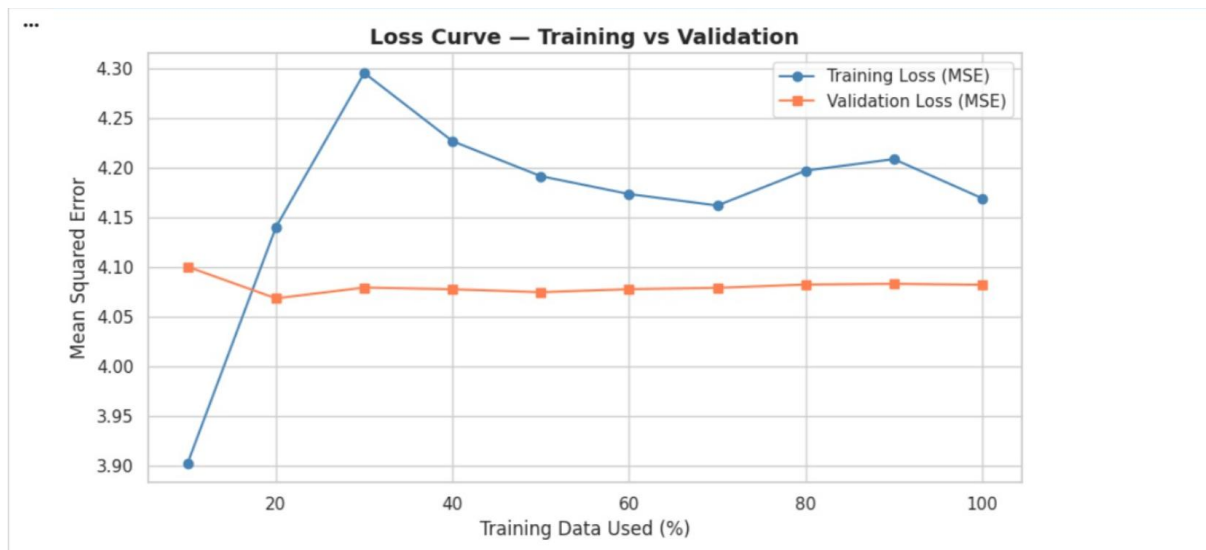


Figure 1: Loss Curve.

```

from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

y_pred = model.predict(X_test_scaled)

mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("==== Model Evaluation ====")
print(f"MAE (Mean Absolute Error) : {mae:.4f}")
print(f"MSE (Mean Squared Error) : {mse:.4f}")
print(f"RMSE (Root Mean Sq. Error) : {rmse:.4f}")
print(f"R2 (R-Squared Score) : {r2:.4f}")

```

```

... ===== Model Evaluation =====
MAE (Mean Absolute Error) : 1.6111
MSE (Mean Squared Error) : 4.0826
RMSE (Root Mean Sq. Error) : 2.0206
R2 (R-Squared Score) : 0.9890

```

Figure 2: Accuracy Metrics.

```

comparison = pd.DataFrame({
    'Actual Performance': y_test.values[:10],
    'Predicted Performance': np.round(y_pred[:10], 2),
    'Error': np.round(y_test.values[:10] - y_pred[:10], 2)
})

print("          Sample Predictions (First 10) ")
print(comparison.to_string(index=False))

```

```

...          Sample Predictions (First 10)
Actual Performance Predicted Performance Error
51.0              54.71 -3.71
20.0              22.62 -2.62
46.0              47.90 -1.90
28.0              31.29 -3.29
41.0              43.00 -2.00
59.0              59.07 -0.07
48.0              45.90  2.10
87.0              86.46  0.54
37.0              37.70 -0.70
73.0              72.06  0.94

```

Figure 3: Sample Predictions.

2. Logistic Regression:

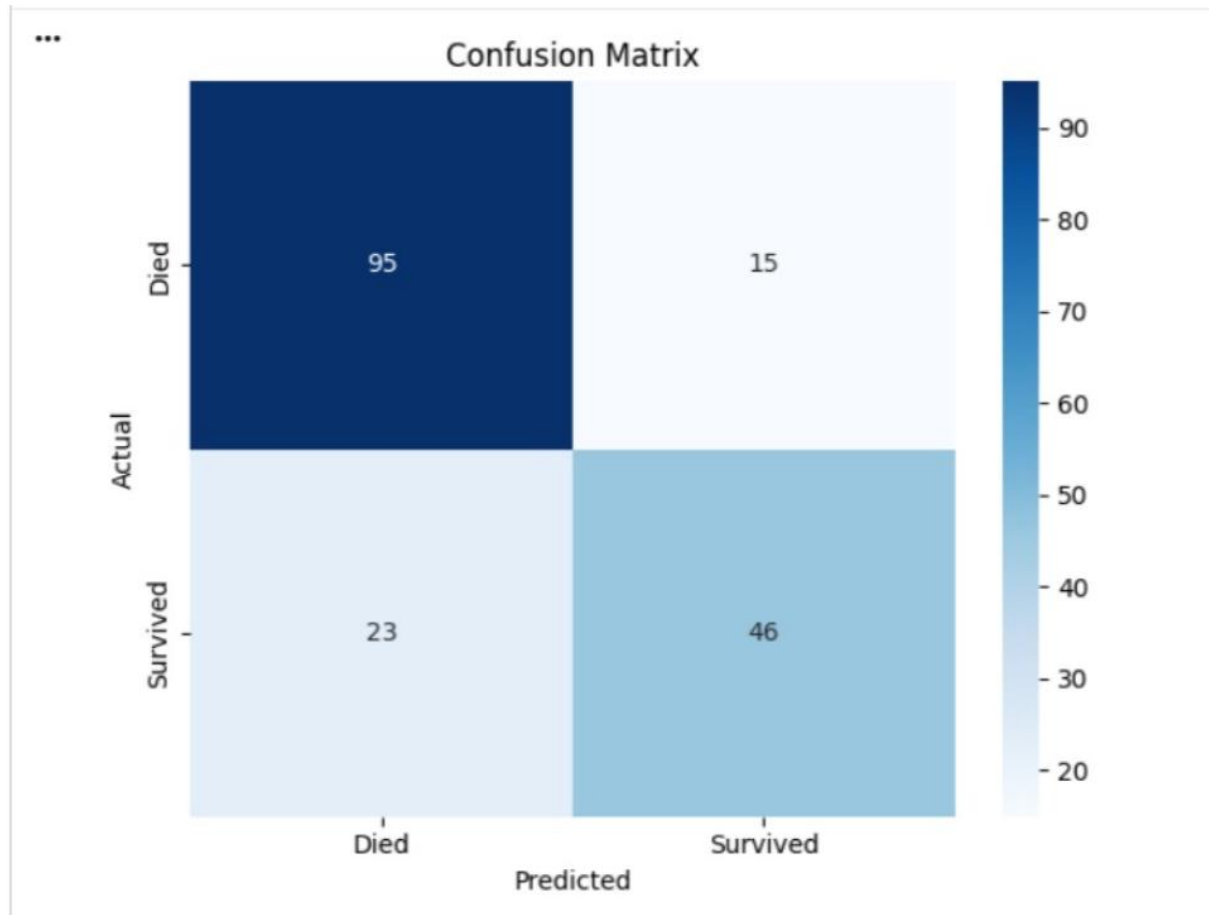


Figure 4: Confusion Matrix.



```
print("          FINAL MODEL SUMMARY")

print(f" Dataset      : Titanic (Kaggle)")
print(f" Model         : Logistic Regression")
print(f" Train size     : {X_train.shape[0]} samples")
print(f" Val size       : {X_val.shape[0]} samples")
print(f" Test size      : {X_test.shape[0]} samples")
print("_" * 45)
print(f" Accuracy       : {accuracy_score(y_test, y_test_pred):.4f}")
print(f" Precision      : {precision_score(y_test, y_test_pred):.4f}")
print(f" Recall         : {recall_score(y_test, y_test_pred):.4f}")
print(f" F1-Score       : {f1_score(y_test, y_test_pred):.4f}")
print(f" AUC-ROC        : {roc_auc_score(y_test, y_test_prob):.4f}")
```

```
...          FINAL MODEL SUMMARY
Dataset      : Titanic (Kaggle)
Model        : Logistic Regression
Train size   : 534 samples
Val size     : 178 samples
Test size    : 179 samples
-----
Accuracy     : 0.7877
Precision    : 0.7541
Recall       : 0.6667
F1-Score     : 0.7077
AUC-ROC      : 0.8474
```

Figure 5: Evaluation Metrics.

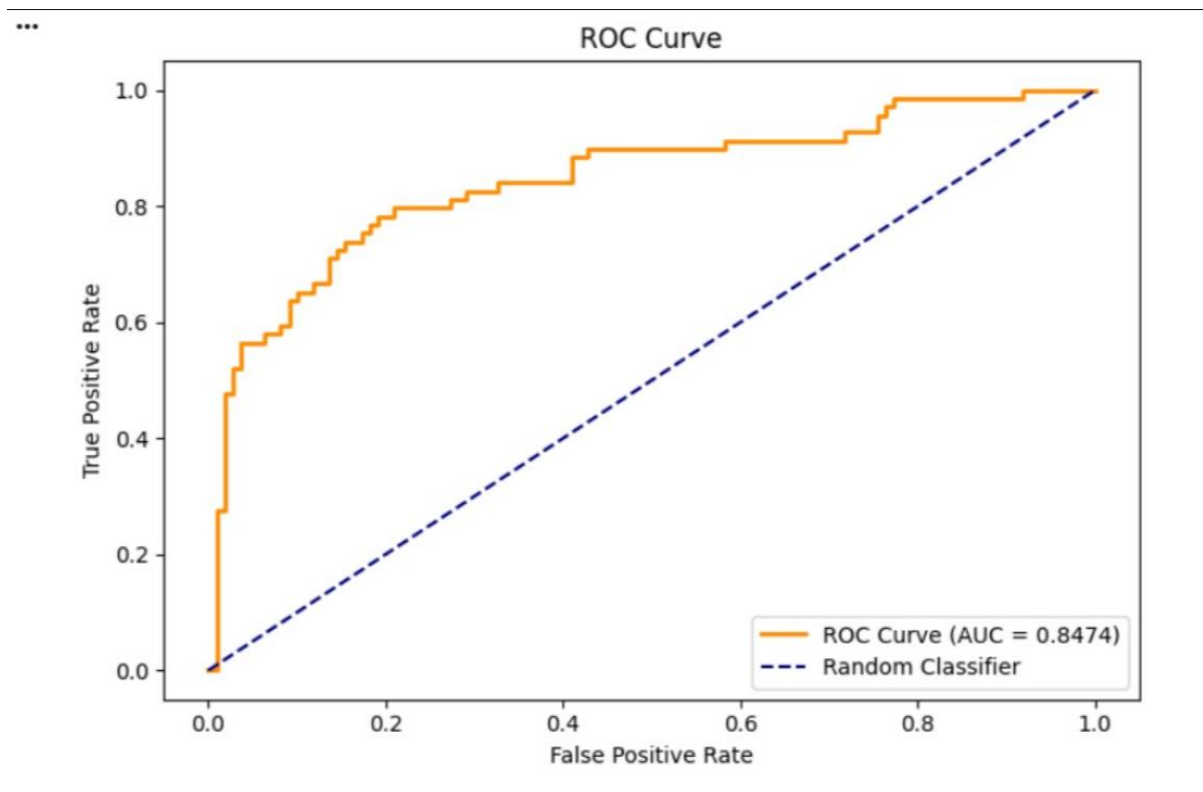


Figure 6: ROC Curve.

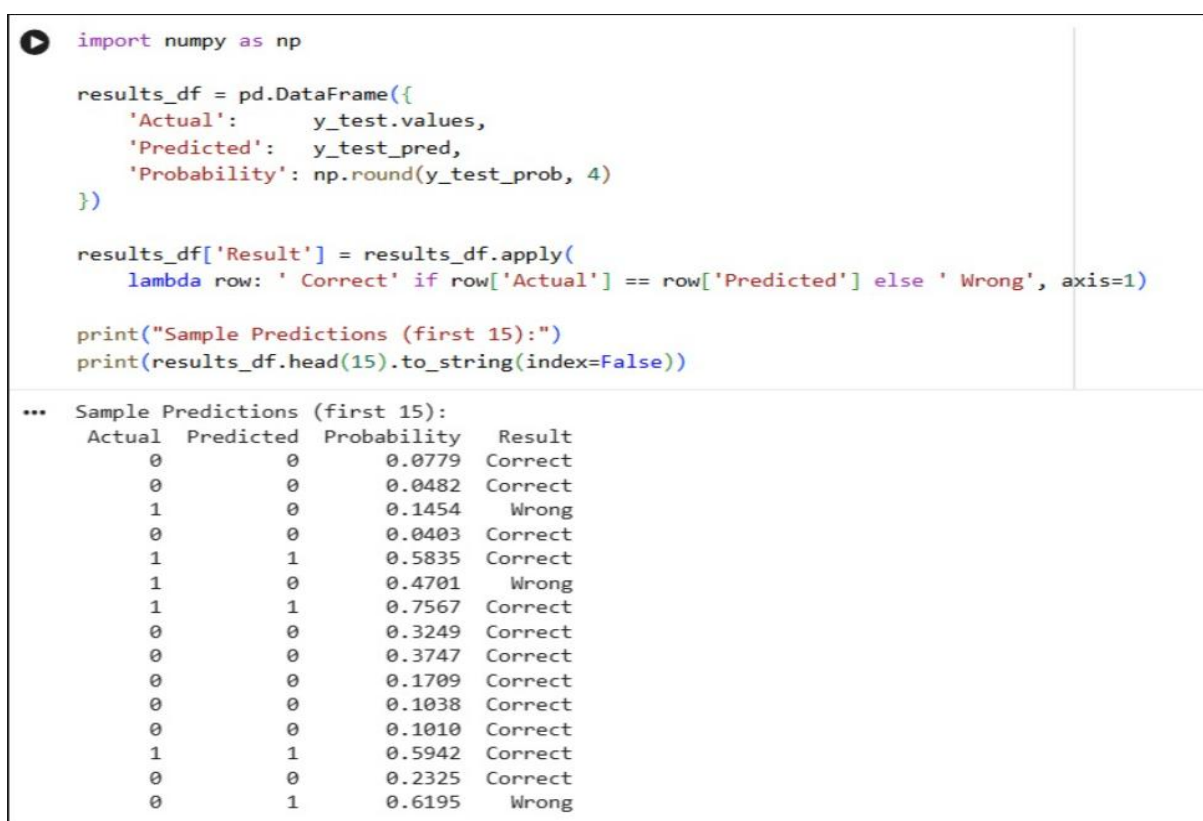


Figure 7: Sample Predictions.

